

Causal associations between plasma proteins and prostate cancer: a Proteome-Wide Mendelian Randomization

Reviewed Preprint

v1 • May 12, 2025

Not revised

Chen Lin, Rong Zhuona, Gu Yanlun, Chen Yuke, Yu Wei, Zhou Ying , Pang Xiaocong 

Department of Pharmacy, Peking University First Hospital, Beijing, China • School of Pharmaceutical Sciences, Peking University, Beijing, China • Department of Urology Surgery, Peking University Third Hospital, Beijing, China

 https://en.wikipedia.org/wiki/Open_access Copyright information

eLife Assessment

This study presents a **valuable** meta-analysis of two independent genome-wide association studies (GWASs) elucidating the role of plasma proteins as biomarkers for improving early detection of prostate cancer (PCa). The evidence supporting novel protein biomarkers of PCa risk is **solid**, although exploration of how these markers may also be shared with other prostate diseases would have strengthened the study. The work will be of interest to the field for elucidating novel variants of prostate cancer risk.

<https://doi.org/10.7554/eLife.101584.1.sa3>

Abstract

Background

Due to the limitations in specificity of current diagnostic methods for prostate cancer (PCa), more reliable biomarkers are needed to explore for improving early detection. Plasma proteins represent a promising source of biomarkers, therefore understanding the causal relationships between specific plasma proteins and PCa could be conducive to identify novel biomarkers and therapeutic targets for PCa prevention and treatment.

Methods

We performed a meta-analysis of two independent genome-wide association studies (GWASs) including 94,397 individuals with PCa and 192,372 controls. A mendelian randomization (MR) supplemented by colocalization analysis was conducted, using *cis*-acting variants on 4,907 plasma proteins from deCODE Genetics (N=35,559) and 2,940 plasma proteins from UK Biobank Pharma Proteomics Project (UKB-PPP) (N=54,219). Then, the biological pathway analysis and druggability evaluation of the risk proteins were further performed.

Results

Five possible susceptibility loci (JAZF1, PDILM5, WDPCP, EEFSEC, and TNS3) for PCa were identified through the meta-analysis of GWASs. Among 3,722 plasma proteins, 193 proteins were associated with PCa risk, of which 20 high-risk proteins, including KLK3, were validated in both the deCODE and UKB-PPP cohorts. Functional annotation of these genes encoding proteins confirmed enrichment of immune response, inflammatory response, cell-cell interaction and so on. Genetic colocalization and druggable genome analyses also identified several potential drug targets for PCa, such as HSPB1, RRM2B and PSCA.

Conclusions

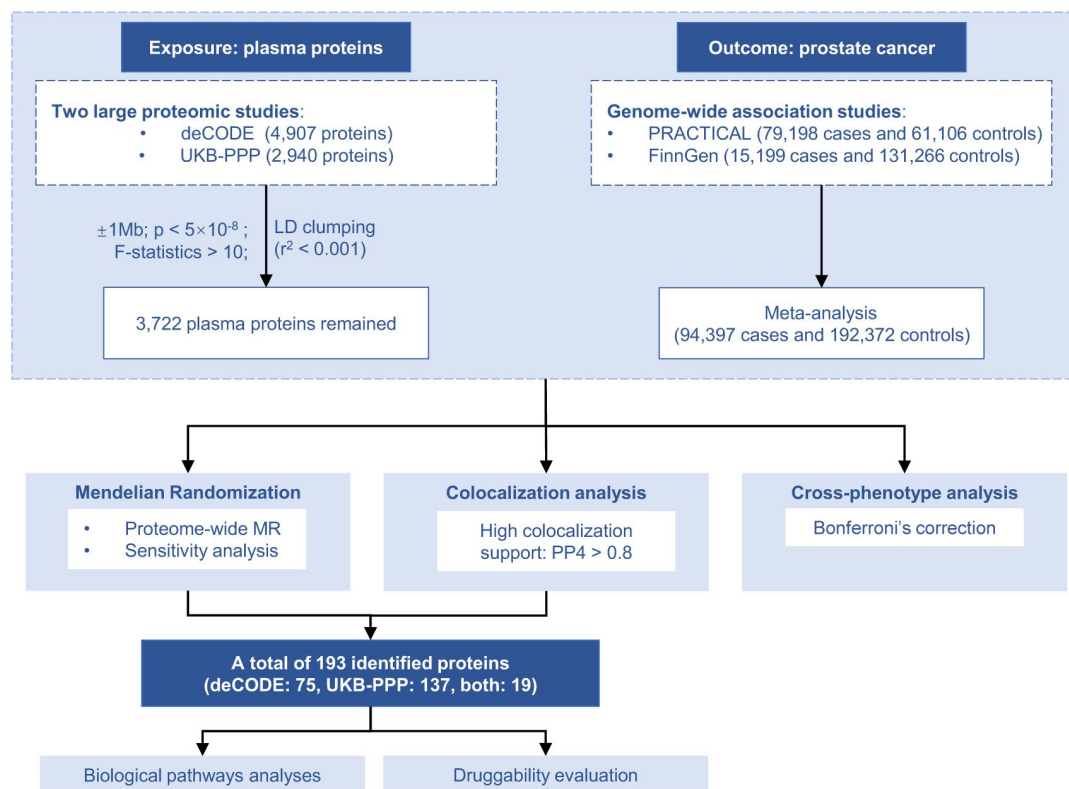
We identified novel variants as well as several protein biomarkers linked to PCa risk and indicated pathways associated with PCa, which offered new insights into PCa etiology and contributed to development of novel biomarkers for early detection and potential therapeutic interventions.

Funding

This work was supported by grants from Beijing Municipal Natural Science Foundation (grant No. JQ24059, No. L234038) and National Natural Science Foundation of China (grant No. 82274015).

Graphical abstract

A comprehensive study design summarized as follows.



Introduction

Prostate cancer (PCa) is the second most common malignancy with an estimation of 1.5 million new cases in 2022, which is account for 14% of total cancer diagnosed in men worldwide (Ferlay *et al.*, 2024). Even more worrying is that the number of new cases of PCa are projected to 2.9 million by 2040 (James *et al.*, 2024). Despite advancements in medical technology, early detection of PCa remains a significant challenge due to the limitations in specificity of current diagnostic methods, such as prostate-specific antigen (PSA) testing (Carter *et al.*, 2013; Draisma *et al.*, 2009; Oesterling, 1991). Therefore, there is an urgent need for more accurate and reliable biomarkers that can improve early detection and prognostic evaluations of PCa. Plasma proteins have key roles in the development and progression of PCa, such as interleukin (IL)-6 (Deichaite *et al.*, 2022), insulin-like growth factor (IGF)-1 and IGF-binding protein (IGFBP)-1 (Cao *et al.*, 2015). However, observational studies exploring the association between plasma proteins and PCa risk are often limited by confounding factors and selection bias, making it challenging to establish a clear causal relationship.

To overcome these limitations, Mendelian randomization (MR) offers a robust methodological approach. This approach utilizes the principle of random assortment of genes from parents to offspring, using genetic variation as an instrumental variable, which mimics the randomization process in a controlled trial, thereby minimizing the impact of reverse causality (Davies, Holmes, & Smith, 2018). With the recent development of proteomics technology, several large-scale proteomic studies have identified over 18,000 protein quantitative trait loci (pQTLs) covering more than 4800 proteins (Feringstad *et al.*, 2021; Pietzner *et al.*, 2021). By using MR, researchers can gain a deeper comprehension of whether particular plasma proteins hold a causal nexus with PCa, thereby potentially unraveling novel biomarkers and targets for the prevention and treatment of PCa.

In this study, we performed a meta-analysis of two genome-wide association studies (GWAS) on PCa (PRACTICAL and FinnGen) for a total sample size of 94,397 cases and 192,372 controls. Based on GWAS summary statistics data from deCODE and UKB-PPP cohorts, we further performed a protein-wide MR (PW-MR) study, supplemented by colocalization analysis, to explore the casual relationship between plasma proteins and PCa risk. Moreover, we indicated biological processes and pathways associated with PCa, and evaluated the druggability of risk proteins. We aimed to identify novel plasma proteins biomarkers for PCa, which could address the limitations of current diagnostic methods and offer new insights into the biological mechanisms of PCa and potential therapeutic targets for intervention.

Results

Meta-analysis of the Genome-Wide Association Studies for Prostate Cancer

We conducted a meta-analysis combining two GWAS studies with a collective sample size of 94,397 individuals with PCa and 192,372 controls, aiming to identify genetic variants linked to PCa. The associations and assessment of SNPs heterogeneity that passed the genome-wide P-value threshold at these loci with PCa were presented in Supplementary Table 1. We found 5 genetic risk loci contained at least one SNP passing the genome-wide significance threshold of $P \leq 5 \times 10^{-8}$: JAZF1, PDILM5, WDPCP, EEFSEC, and TNS3 (Figure 1). Among them, PDLIM5, WDPCP, EEFSEC and TNS3 were promising candidates as novel susceptibility loci associated with PCa. Supplementary Figure 1 displayed the associated quantile-quantile plot. The LocusZoom plots of the top SNPs at JAZF1, PDILM5, WDPCP, EEFSEC and TNS3, along with their genomic location, GWAS P values and recombination rate with neighboring SNPs were visualized in Supplementary Figure 2. In summary, this GWAS meta-analysis discovered genetic variations in one recognized PC-associated loci and four potential novel loci, providing a reliable dataset for MR analyses.

Cross-phenotype Analysis of Prostate Cancer

We used the iCPAGdb to conduct the cross-phenotype genetic association analyses with PCa genome-wide significant SNPs ($P \leq 5 \times 10^{-8}$). The iCPAGdb offered an improved algorithm for identifying cross-phenotype associations by using pre-computed ancestry-specific LD databases and integrating genetic data from 3793 traits in the NHGRI-EBI GWAS catalog (L. Wang et al., 2020). After adjusting for Bonferroni's correction, the cross-phenotype analysis indicated that 117 traits showed significant association with PCa (Supplementary Table 2). As shown in Figure 2, the top 10 enrichments were for prostate specific antigen measurement ($P=7.16 \times 10^{-43}$), body height ($P=2.18 \times 10^{-30}$), BMI-adjusted waist-hip ratio ($P=1.32 \times 10^{-26}$), high density lipoprotein cholesterol measurement ($P=2.38 \times 10^{-22}$), systolic blood pressure ($P=2.93 \times 10^{-22}$), balding measurement ($P=3.01 \times 10^{-22}$), leukocyte count ($P=3.46 \times 10^{-21}$), blood protein measurement ($P=4.95 \times 10^{-21}$), body mass index ($P=1.71 \times 10^{-20}$) and heel bone mineral density ($P=2.02 \times 10^{-19}$).

Proteome-Wide Mendelian Randomization Studies of Prostate Cancer

The genetic association summary statistics of 35,559 Icelanders from deCODE Genetics and 54,219 Europeans from the UKB-PPP were utilized to investigate the relationship between PCa and plasma proteins. Our genetic instrument selection strategy enabled us to examine 1778 and 1944 proteins from deCODE and UKB-PPP (Supplementary Table 3A and B). Using the Wald ratio or IVW method, a total of 193 unique plasma proteins were significantly associated with PCa after multiple tests with a 5% FDR correction. This analysis yielded 137 proteins in UKB-PPP more than

Figure 1.

Manhattan plot of PCa GWAS meta-analysis.

The genetic regions containing top SNPs related to PCa are depicted. The red dashed line signifies the genome-wide significance threshold of 5.0×10^{-8} .

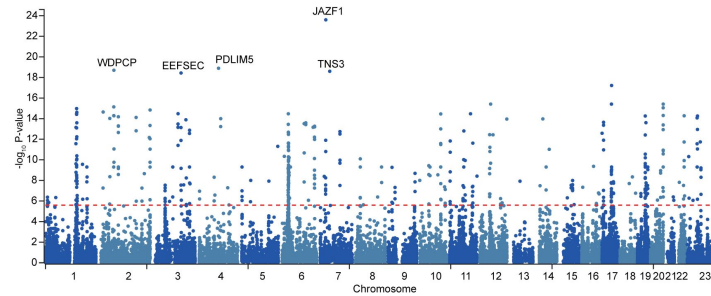
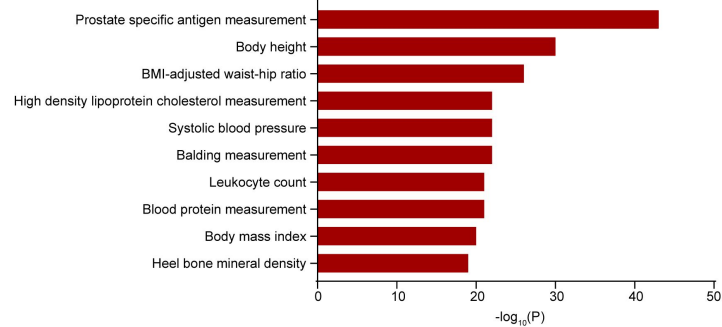


Figure 2.

The top 10 significant cross-phenotype associations with PCa at a 5% FDR.

The x-axis represents the P value of the correlation. BMI, body mass index.



76 proteins in deCODE (**Figure 3A** [↗](#)). The results of Heterogeneity test based on Q statistics showed little evidence of heterogeneity. Furthermore, no significant intercept was detected, implying that there was no directional pleiotropy observed.

After FDR correction, 20 proteins were detected in both data sets and the identified associations were consistent (**Figure 3B** [↗](#)). Genetic prediction indicated that 11 of these 20 proteins were positively associated with the risk of PCa as well as remaining 9 proteins were negatively associated, suggesting that these 9 proteins maybe protective factors against PCa (Supplementary Table 4). A PhenoGram depicted the chromosomal location of the 193 unique identified proteins in deCODE and UKB-PPP studies (**Figure 3C** [↗](#)).

Colocalization analysis

We performed colocalization analyses of proteins significantly expressed in deCODE and UKB-PPP studies with PCa. It was observed that 4 proteins in deCODE study and 7 proteins in UKB-PPP were colocalized with PCa associations with high support of evidence ($PPH4 \geq 0.8$) (**Table 1** [↗](#)), suggesting that these 10 plasma proteins might serve as potential targets for treating PCa. Among them, SERPINA3 showed strong colocalization evidence in both deCODE ($PPH4=0.952$) and UKB-PPP ($PPH4=0.951$) studies. This analysis identified 1 causal variant in deCODE (rs61976125) and 1 causal variant in Fenland (rs6575449) (**Figure 4** [↗](#)).

Gene-Based Association and Pathway Analyses

We used the GENE2FUNC tool available in FUMA to explore the biological significance, functional implications and tissue-specific expression of the genes identified from our GWAS. These genes appeared to be significantly enriched in inflammatory and immune pathways (such as defense, immune and inflammatory response), as well as cell interaction and signaling pathways (such as interaction between organism and cell adhesion) (**Figure 5A** [↗](#)). The KEGG pathway enrichment analysis revealed several pathways that were significantly enriched, such as cytokine-cytokine receptor interaction, p53 signaling pathway, JAK-STAT signaling pathway, pathways in cancer and apoptosis (**Figure 5B** [↗](#)). These analyses indicated potential biological processes and mechanisms associated with PCa. The 193 unique genes were predominantly expressed in the lymphocytes, blood, liver and prostate (Supplementary Figure 3).

Druggability of identified proteins

Exploring new therapeutic opportunities for PCa based on genetic information is crucial for developing targeted treatments. In our study, we examined the druggability of genes and proteins identified in MR analyses using OpenTargets databases. We investigated 128 unique drugs targeting 45 identified proteins (Supplementary Table 5). Among these, three drugs (APATORSEN, TRIAPINE and MK-4721) currently in clinical trials were intended for PCa treatment, with each targeting HSPB1, RRM2B and PSCA, respectively. Notably, the effects of these drugs on their respective protein targets align with the directions indicated by our MR results, suggesting a consistency between genetic evidence and therapeutic potential. In addition, several other identified targets, such as RET, FGFR3, NCAM1, TYMP, TNFRSF10B, MMP3, TACSTD2 and NOTCH2, were implicated in various cancers and present potential therapeutic avenues.

Discussion

In this study, we conducted a comprehensive meta-analysis of two GWAS for PCa, identifying significant genetic loci associated with PCa risk. The combined sample size of 94,397 cases and 192,372 controls revealed one known (JAZF1) and four potential novel (PDLIM5, WDPCP, EEFSEC

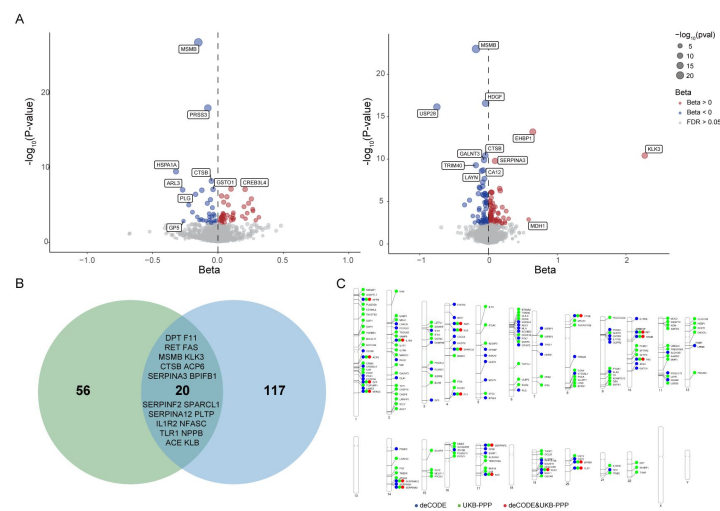


Figure 3.

Result of PW-MR on the associations between plasma proteins and the risk of PCa.

(A) Volcano plot of PCa PW-MR study using deCODE (the left side) and UKB-PPP (the right side) cohorts. Annotated proteins passed the 5% FDR IVW P-value threshold. The blue and red colors represent a negative and positive effect, respectively. (B) Venn diagram depicting proteins associated with PCa in deCODE and UKB-PPP. (C) PhenoGram of PCa PW-MR study significant associations. The blue dots and the green dots represent the deCODE and UKB-PPP specific proteins, respectively. The red dot represents both simultaneously.

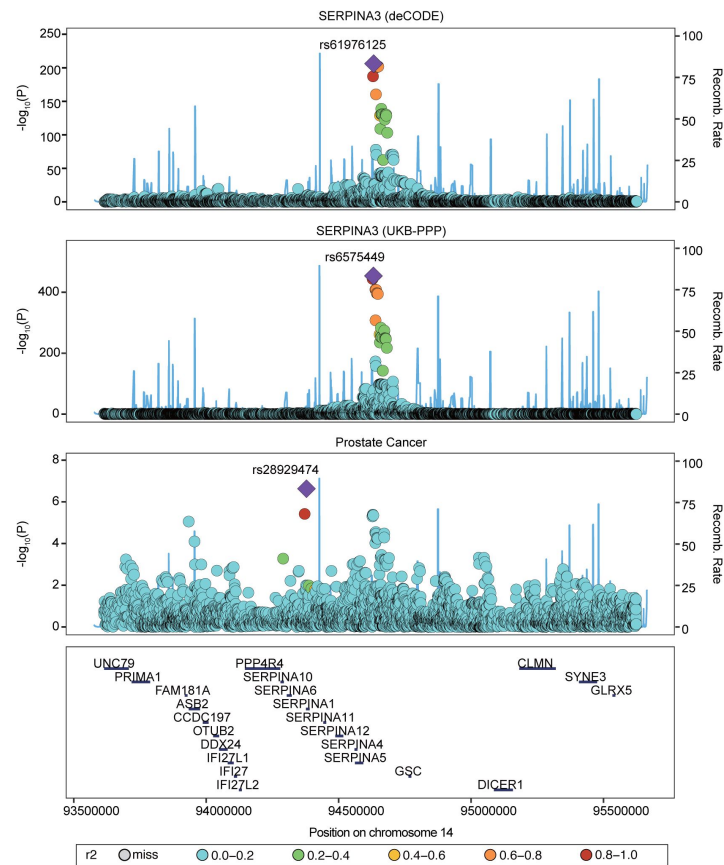


Figure 4.

Colocalization plot of SERPINA3 variants associated with PCa in deCODE and UKB-PPP.

Variants are color-coded based on their linkage disequilibrium (LD) with the lead SNP (the variant with the lowest p-value). The lead SNP is highlighted in red. Other variants are colored from blue to yellow, indicating decreasing LD with the lead SNP.

Table 1.

Analysis of Mendelian randomization and colocalization of significant proteins with PCa.

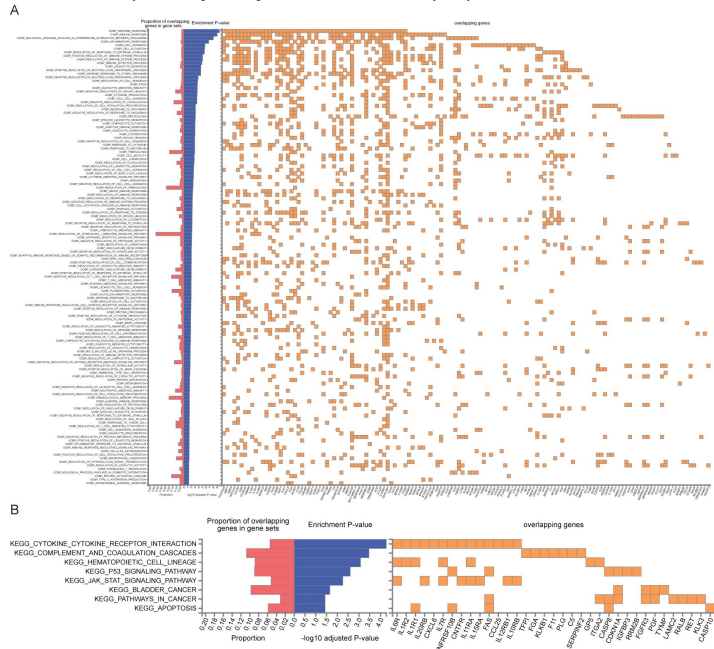
GWAS	Outcomes	Proteins	Mendelian randomization			Colocalization Analysis PH4
			OR (95%CI)	P value	P value after FDR adjustment	
deCODE	Prostate Cancer	MSMB	0.86(0.84,0.88)	1.56E-27	2.82E-24	0.999
		POGLUT3	0.89(0.83,0.96)	1.16E-03	2.44E-02	0.998
		PRSS3	0.93(0.91,0.94)	1.00E-18	6.03E-16	0.956
		SERPINA3	1.11(1.06,1.15)	7.18E-08	1.05E-05	0.952
UKB-PPP	Prostate Cancer	USP28	0.47(0.40,0.56)	7.27E-17	2.89E-14	0.999
		KLK3	9.70(4.94,19.03)	3.83E-11	6.91E-09	0.999
		IGFBP3	1.07(1.05,1.09)	2.04E-11	4.49E-09	0.997
		CASP10	1.22(1.11,1.34)	4.73E-05	1.54E-03	0.988
		HDGF	0.96(0.95,0.97)	2.61E-17	1.29E-14	0.966
		SERPINA3	1.10(1.07,1.13)	1.64E-10	2.32E-08	0.951
		C5	1.19(1.11,1.27)	5.32E-07	4.06E-05	0.883

OR, odds ratio; CI, confidence interval; FDR, false discovery rate.

Figure 5.

Functional annotation of the genetic architecture of PCa.

(A) Biological processes and (B) KEGG pathway analysis of the 193 unique proteins identified in deCODE and UKB-PPP.



and TNS3) susceptibility loci of PCa. JAZF1 (Juxtaposed with another zinc finger gene 1) is a transcriptional repressor of testicular nuclear receptor 4 (TR4) (Nakajima, Fujino, Nakanishi, Kim, & Jetten, 2004). Several GWAS studies have implicated that JAZF1 is highly associated with type 2 diabetes and PCa risk (Frayling, Colhoun, & Florez, 2008; Machiela et al., 2012; Saxena, Voight, Zeggini, Scott, & Genetics, 2008; Thomas et al., 2008). JAZF1 may influence the risk of PCa through its role in metabolic regulation (Rosario et al., 2023) and cellular proliferation (Sung et al., 2018).

PDLIM5 (PDZ and LIM domain 5) is a member of the *enigma* subfamily and features a N-terminal PDZ domain and three C-terminal LIM domains (X. J. Wang & Su, 2010). A study implicated that downregulating the expression of PDLIM5 could ultimately impede the progression of PCa (Xie et al., 2020). WDPCP, a PCP effector, can regulate PCP by directly modulating the actin cytoskeleton (Cui et al., 2013). Disruption of PCP signaling can cause abnormal cell behavior and tissue structure, leading to the development of cancer (Humphries & Mlodzik, 2018). EEFSEC (eukaryotic elongation factor, selenocysteine-tRNA specific) was identified as a genome-wide significant loci in recent chronic obstructive pulmonary disease GWAS (Hobbs et al., 2017; Wain et al., 2017). TNS3 (Tensin 3) is involved in cell migration, invasion and adhesion processes (Chen et al., 2017; Martuszczyńska et al., 2009; Zheng et al., 2021; Zuidema et al., 2022), which are critical in cancer metastasis. To certain whether these 4 genes causative factors for PCa, genetic association studies with a larger sample size will be needed.

In the cross-phenotype Analysis of PCa, we found that prostate specific antigen (PSA) measurement, BMI-adjusted waist-hip ratio (WHR), high density lipoprotein cholesterol (HDL) measurement, systolic blood pressure, balding measurement and other traits were significantly associated with PCa. These findings highlighted the multifactorial nature of PCa and underscored the importance of considering a broad range of physiological and biochemical factors in understanding its etiology and progression.

PSA measurement, a protein secreted by both normal and malignant prostate epithelial cells, has long been a cornerstone in the early detection and monitoring of PCa (Oesterling, 1991). However, the specificity and sensitivity of PSA as a diagnostic tool have been questioned due to its elevation in benign conditions such as prostatitis and benign prostatic hyperplasia, leading to unnecessary biopsies (Carter et al., 2013) and substantial overdiagnosis (Draisma et al., 2009). Therefore, supplementary biomarkers are urgently needed to improve diagnostic accuracy.

Obesity has emerged as a significant risk factor in cancer development (Petrelli et al., 2021), such as aggressive PCa (Moyad, 2015). The significant association between BMI-adjusted WHR and PCa suggested that central obesity may play a critical role in the development of PCa (Perez-Cornago, Dunneram, Watts, Key, & Travis, 2022).

Central adiposity, primarily referring to visceral adipose tissue, can influence hormone levels and inflammatory processes, thereby contributing to carcinogenesis (Doyle, Donohoe, Lysaght, & Reynolds, 2012). HDL cholesterol is known for its atherosclerotic protective effects (Rohatgi, Westertorp, von Eckardstein, Remaley, & Rye, 2021), mainly due to the ability to promote the reverse transport of cholesterol from peripheral cells to the liver for excretion (Cuchel & Rader, 2006), and to exert antioxidant and anti-inflammatory activities. Through similar mechanisms, HDL may influence the development and progression of tumors, by directly interacting with cancer cells or by modifying the tumor microenvironment (Ossoli, Wolska, Remaley, & Gomaraschi, 2022). Higher HDL levels might confer a protective effect against PCa by influencing cancer cells proliferation, possibly through mechanisms involving antioxidant (Ruscica et al., 2018) and anti-inflammatory properties.

Results from previous studies of the association between hypertension and PCa development have been inconsistent(Christakoudi et al., 2020 [↗](#); Liang et al., 2016 [↗](#); Seretis et al., 2019 [↗](#)). One MR analysis suggested that elevated systolic blood pressure might be linked to an increased risk of PCa, potentially due to systemic inflammation(Stikbakke et al., 2022 [↗](#)), which contributed to a pro-tumorigenic environment by promoting cellular proliferation, DNA damage, and resistance to apoptosis(Mantovani, Allavena, Sica, & Balkwill, 2008 [↗](#)). However, the precise relationship between systolic blood pressure and PCa remains unclear. More research is needed to elucidate the exact biological pathways involved and to determine whether managing hypertension could serve as a preventive strategy for PCa.

The underlying mechanism for the link between androgenic alopecia and an increased risk of PCa may involve androgen metabolism, as both conditions are influenced by dihydrotestosterone (DHT) levels. Elevated DHT can promote the miniaturization of hair follicle leading to balding(Kaufman, 1996 [↗](#)), as well as stimulate prostate cell proliferation(Tong et al., 2022 [↗](#)), contributing to the development of PCa. While these associations highlight a potential shared hormonal pathway, further studies are needed to confirm the causality. Overall, our cross-phenotype analysis provided a comprehensive view of the diverse factors associated with PCa and enhanced our understanding of the disease's multifaceted nature. Future researches should focus on elucidating the underlying mechanisms of these associations and exploring their potential for integration into clinical practice.

We performed a PW-MR study supplemented by colocalization analysis to explore the casual association of 3,722 plasma proteins with the risk of PCa. MR analysis identified a total of 193 unique proteins significantly associated with PCa, of which 10 proteins had a strong support of colocalization. Microseminoprotein-beta (MSMB), also known as PSP94 or beta-inhibin, as an immunoglobulin superfamily protein and one of the most abundant proteins secreted by prostate epithelial cells has the strongest correlation(Byrne et al., 2019 [↗](#)). We predicted that men with lower blood levels of MSMB have higher risk of PCa, based on two European ancestry cancer GWAS studies. These results were consistent with another published prospective study(Haiman et al., 2013 [↗](#)), supporting a potential protective role of MSMB in the development of PCa. One previous publication reported that the decreased expression of MSMB was both in the tumor (especially in more advanced tumor) and adjacent benign prostate tissue(Bergström, Järemo, Nilsson, Adamo, & Bergh, 2018 [↗](#)), which is not in conflict as it may be related to aggressiveness of prostate tumors. Although the mechanism of action of MSMB in PCa is unclear, it has been shown to control prostate cell growth by regulating apoptosis(Garde et al., 1999 [↗](#)). In addition to our study, several other studies also supported the potential of MSMB as a serum marker for the early detection of PCa(Nam et al., 2006 [↗](#); Reeves, Dulude, Panchal, Daigneault, & Ramnani, 2006 [↗](#)). Consequently, further research is essential to explore the functional role and possible clinical utility of MSMB in PCa.

Serine protease inhibitor A3 (SERPINA3), an inhibitors of serine proteases that promotes tumor development by regulating the transcription of some oncogenes and its elevation was associated with a worse prognosis in some cancers(De Mezer et al., 2023 [↗](#)). SERPINA3 was up-regulated in PCa cells and enhanced cells migration and invasion(Z. S. Xing, Li, Liu, Zhang, & Bai, 2021 [↗](#)), which was consistent with our prediction of it as a risk factor in this MR Study. A study identified that the enhanced expression of SERPINA3 stimulated the bone environment, therefore it may be served as a diagnostic biomarker to predict PCa with bone metastasis phenotype and survival(Ito et al., 2023 [↗](#)).

Protease serine 3 (PRSS3), also named mesotrypsin, is reported to be aberrantly expressed in various types of tumors and participates in the progression and development of cancers. For example, PRSS3 was identified to be downregulated in lung cancer(Zhou, Li, & Luo, 2023 [↗](#)) but upregulated in pancreatic cancer(Jiang et al., 2010 [↗](#); C. J. Xing et al., 2019 [↗](#)), suggesting it may have different roles depending on the cellular or disease ways. A study of *vivo* and *vitro*

experiments has demonstrated that PRSS3 played an important role in PCa metastasis (Hockla et al., 2012). Despite these findings, further studies are necessary to fully understand the functions and mechanisms of PRSS3 in human PCa.

KLK3 (Kallikrein Related Peptidase 3), also known as PSA, is a serine protease predominantly expressed in the prostate gland. One potential mechanism through which KLK3 may influence PCa development involved its proteolytic activity, which can degrade extracellular matrix components and facilitate tumor invasion and metastasis (Avgeris, Mavridis, & Scorilas, 2012; Lawrence, Lai, & Clements, 2010; Mavridis, Avgeris, & Scorilas, 2014). Additionally, the androgen regulation of KLK3 expression correlated with the androgen receptor signaling pathway, a critical driver of PCa progression, particularly in castration-resistant prostate cancer (CRPC) (Lilja, Ulmert, & Vickers, 2008). Despite these insights, the precise biological pathways through which KLK3 contributes to PCa progression remain incompletely understood. Further research is needed to illustrate the direct and indirect effects of KLK3 on PCa cells and the tumor microenvironment.

Lastly, the druggability evaluation identified 45 protein biomarkers that have been targeted by 128 drugs. Among these, three drugs targeting HSPB1, RRM2B and PSCA, respectively, are currently in clinical trials for potential use in PCa treatment. HSPB1, also known as human HSP27, is a small heat-shock protein that functions as a molecular chaperone, protecting cells against stress-induced damage and apoptosis (Okuno, Adachi, Kozawa, Shimizu, & Yasuda, 2016). It was overexpressed in various cancers (Sheng et al., 2017; Shi et al., 2019), including PCa (Shiota et al., 2013; Vasiljevic et al., 2013), and was associated with tumor progression, metastasis and resistance to therapy. Evidences *in vitro* and *in vivo* suggested that inhibiting HSP27 inhibits tumor growth and sensitivity to cytotoxic chemotherapy (Hadaschik et al., 2008; Kamada et al., 2007). APATORSEN is an antisense oligonucleotide designed to inhibit HSPB1 expression (Jansen & Zangemeister-Wittke, 2002), which is still in clinical trials.

RRM2B is a subunit of ribonucleotide reductase, an enzyme critical for DNA synthesis and repair (Okumura et al., 2005). Its expression was upregulated in response to DNA damage and was involved in maintaining genomic stability (Aye, Li, Long, & Weiss, 2015; Foskolou et al., 2017). Elevated levels and distinct mutation signatures of RRM2B have been linked to poor prognosis in several cancers (Iqbal et al., 2021). PSCA is a cell surface glycoprotein highly expressed in high-grade PCa (Gu et al., 2000; Zhigang & Wenlv, 2004) and other solid tumors (Teng et al., 2022), and was involved in cell proliferation, adhesion and survival (Li et al., 2017).

This study possessed several strengths, including the use of the largest collection of plasma protein data (covering more than 4800 proteins), large sample sizes of GWAS studies, a mutual validation across two independent outcome datasets and the use of colocalization analysis to support the MR results. Additionally, our assessment of the human blood proteome relied on two technologies (SOMAmer and Olink), which were valuable for identifying plasma proteins associated with diseases traits such as PCa.

Some limitations of our analysis should be acknowledged. Firstly, this investigation was only confined to Europeans, restricting the applicability of our findings to other populations. It is crucial for future studies to identify risk proteins in more diverse populations, especially African ancestry who face a higher risk of PCa. Moreover, our study concentrated on the proteins available in the MR analysis, which likely led to the omission of other potential therapeutic targets. Lastly, we utilized proteomic data from Icelanders whose genetic backgrounds may differ from other European populations, which might introduce bias. However, this potential bias might be minimal, as we found that 20 proteins showing significance ($P < 0.05$) in both the deCODE and UKB-PPP studies had consistent associations with PCa.

In conclusion, we conducted a large-scale PW-MR study using the MR and colocalization analysis to investigate the genetic associations of up to 3,722 unique proteins with PCa. We revealed the complex genetic architecture of PCa and identified many new plasma proteins with strong causal associations to PCa. These findings highlighted the potential biomarkers for early detection and therapeutic targets, providing a foundation for future research and potential clinical applications.

Methods

Data sources for plasma proteins

We selected *cis*-SNPs associated with plasma proteins as instrumental variables from two large-scale GWASs in the deCODE Genetics (Ferkingstad et al., 2021 [↗](#)) and UKB-PPP (Sun et al., 2023 [↗](#)). *Cis*-SNPs were defined as SNPs within a vicinity of ± 1 Mb around the gene encoding the protein. From these SNPs, only those with a minor allele frequency of $\geq 1\%$ that were genome-wide significance ($p < 5 \times 10^{-8}$) and considered independent (linkage disequilibrium $r^2 < 0.1$ in 1000G) were retained. deCODE Genetics conducted proteomic profiling on blood plasma samples from 35,559 Icelanders using the SomaScan platform and collected data on 4907 aptamers (Ferkingstad et al., 2021 [↗](#)). For the two-sample MR analysis, we selected *cis*-SNPs as instrumental variables for 1778 proteins. Likewise, *cis*-SNPs for 1944 plasma proteins were obtained from the UKB-PPP where 2940 proteins were measured among 54,219 Europeans using the Olink platform (Sun et al., 2023 [↗](#)). The proteins with positive MR results were included in the colocalization analysis.

Data sources for Prostate Cancer

GWAS summary statistics for PCa were obtained from the PRACTICAL consortium (Schumacher et al., 2018 [↗](#)) and FinnGen study. The PRACTICAL (Prostate Cancer Association Group to Investigate Cancer Associated Alterations in the Genome) consortium included 79,198 cases and 61,106 controls. We used the data on PCa from the FinnGen study R10 in this analysis, which comprised 15,199 cases and 131,266 controls. All participants of these two cohorts were of European ancestry. In the MR analysis, we treated the PRACTICAL consortium as the discovery study and the FinnGen R10 study as the replication. To increase the effectiveness, we performed a fixed-effect GWAS meta-analysis of the two GWASs using the METAL package. The quantile-quantile plot was generated using the “qqman” package in R software. The R package “gassocplot” was used to plot regional association plots for the top SNP at each of the genome-wide significant loci identified.

Cross-phenotype analysis

Through the interactive Cross-Phenotype Analysis of GWAS database (iCPAGdb) (L. Wang et al., 2020 [↗](#)), a new platform for cross-phenotype analysis, we explored the genetic correlation between PCa and other traits. We analyzed the genetic correlations between PCa and 3793 traits using data from the National Human Genome Research Institute-European Bioinformatics Institute (NHGRI-EBI) GWAS catalog (L. Wang et al., 2020 [↗](#)). Through the use of ancestry LD-specific association data, iCPAGdb performs cross-phenotype enrichment analyses. iCPAGdb reveals signals of pairwise traits and shared signals by analyzing traits associations with LD proxy SNPs. The output data will show results from Fisher’s exact test with adjustment for 5% false discovery rate (FDR) and also Bonferroni’s, Jaccard’s, Sorensen’s, and Chao-Sorensen similarity indexes.

MR analysis

We conducted a MR analysis with plasma proteins as the exposure variable and PC as the outcome variable. We utilized the R package “TwoSampleMR” in R software (4.3.3) for the MR analysis. When only one SNP was available for a particular protein, we applied the Wald ratio test. Inverse-variance weighted (IVW) was used as the main analysis method for two or more SNPs available.

Considering multiple testing, we employed the 5% False Discovery Rate (FDR) method for P-value correction. $P < 0.05$ was considered statistically significant. MR-Egger and weighted median methods were used as supplementary analysis methods. Cochran's Q statistic test using the MR-Egger method was used to determine the heterogeneity between the genetic variants. We also performed MR-Egger regression intercept to detect and adjust the directional horizontal pleiotropy (Hemani, Bowden, & Smith, 2018 [↗](#)), $P < 0.05$ was considered as the presence of directional pleiotropy and thus removed from the further analyses.

Colocalization analysis

The “coloc” package was used to perform Bayesian colocalization analysis to investigate if the associations between plasma proteins and PCa were driven by linkage disequilibrium (LD) (Giambartolomei et al., 2014 [↗](#)). For proteins with positive MR results, the Bayesian method assessed the support for the following five exclusive hypotheses: 1) no association with either trait; 2) association with trait 1 only; 3) association with trait 2 only; 4) both traits are associated, but distinct causal variants were for two traits; and 5) both traits are associated, and the same shares causal variant for both traits (Foley et al., 2021 [↗](#)). The posterior probability is provided for each hypothesis (H0, H1, H2, H3, and H4). In this analysis, we set prior probabilities of the SNP being associated with trait 1 only (p1) at 1×10^{-4} ; the probability of the SNP being associated with trait 2 only (p2) at 1×10^{-4} ; and the probability of the SNP being associated with both traits (p12) at 1×10^{-5} . Two signals were considered to have strong evidence of colocalization if the posterior probability for shared causal variants (PH4) was ≥ 0.8 . The analysis was performed in R software (4.3.3).

Biological pathway analysis

The GENE2FUNC tool in FUMA (Watanabe, Taskesen, van Bochoven, & Posthuma, 2017 [↗](#)) web application programming interface version 1.5.2 were used to investigate functional annotation and enrichment analysis of the genes coding for the 193 unique proteins identified by the PW-MR approach. This analysis involved calculating the log2 fold-change expression for each gene across 54 tissue types from the GTEx (Genotype-Tissue Expression) database. We conducted gene enrichment analysis to identify overrepresented biological processes using the Gene Ontology (GO) database, which helped determine the biological processes most relevant to the genes identified. To investigate the involvement of the identified genes in functional and signaling pathways, we used the KEGG (Kyoto Encyclopedia of Genes and Genomes) pathway database. Enrichment results were considered significant if they passed 5% FDR P-value threshold. Additionally, only enrichments involving at least two overlapping genes with the gene sets were considered to ensure robustness and biological relevance.

Druggability evaluation

The analysis of drug targets for the significant proteins identified in the MR Analysis was conducted using data from OpenTargets (Ochoa et al., 2023 [↗](#)) (v.22.11), which is publicly accessible. We selected all drugs for which there was evidence for an association with the protein of interest.

Data availability

The GWAS summary statistics of blood pQTLs from deCODE can be found at <https://www.decode.com/summarydata/> [↗](#), and those of the UKB-PPP can be found at <https://www.synapse.org/#!/Synapse:syn51365303> [↗](#). Summary statistics of PCa from the PRACTICAL consortium can be obtained from OpenGWAS (<https://gwas.mrcieu.ac.uk/> [↗](#)), and those of the FinnGen study can be obtained from <https://r10.finnngen.fi/> [↗](#). The OpenTargets data were downloaded directly from the OpenTargets website (<https://platform.opentargets.org/downloads/data> [↗](#)).

Acknowledgements

The authors would like to thank all the participants and investigators of the studies (PRACTICAL, FinnGen, deCODE, UKB-PPP) for providing the invaluable data used in this research. Their commitment and effort have been essential for the success of this study. Furthermore, special thanks to our colleagues and collaborators for their insightful discussions and feedback throughout the research process. This research was funded by Beijing Municipal Natural Science Foundation (grant No. JQ24059, No. L234038), and National Natural Science Foundation of China (grant No. 82274015).

Additional files

Supplementary figures [↗](#)

Supplementary tables [↗](#)

References

- Avgeris M., Mavridis K., Scorilas A (2012) **Kallikrein-related peptidases in prostate, breast, and ovarian cancers: from pathobiology to clinical relevance** *Biol Chem* **393**:301–317 <https://doi.org/10.1515/hsz-2011-0260>
- Aye Y., Li M., Long M. J. C., Weiss R. S (2015) **Ribonucleotide reductase and cancer: biological mechanisms and targeted therapies** *Oncogene* **34**:2011–2021 <https://doi.org/10.1038/onc.2014.155>
- Bergström S. H., Järemo H., Nilsson M., Adamo H. H., Bergh A (2018) **Prostate tumors downregulate microseminoprotein-beta (MSMB) in the surrounding benign prostate epithelium and this response is associated with tumor aggressiveness** *Prostate* **78**:257–265 <https://doi.org/10.1002/pros.23466>
- Byrne K. S., Appleby P. N., Key T. J., Holmes M. V., Fensom G. K., Agudo A., ..., Consortium P. (2019) **The role of plasma microseminoprotein-beta in prostate cancer: an observational nested case-control and Mendelian randomization study in the European prospective investigation into cancer and nutrition** *Annals of Oncology* **30**:983–989 <https://doi.org/10.1093/annonc/mdz121>
- Cao Y., Nimptsch K., Shui I. M., Platz E. A., Wu K., Pollak M. N., ..., Giovannucci E. L (2015) **Prediagnostic plasma IGFBP-1, IGF-1 and risk of prostate cancer** *International Journal of Cancer* **136**:2418–2426 <https://doi.org/10.1002/ijc.29295>
- Carter H. B., Albertsen P. C., Barry M. J., Etzioni R., Freedland S. J., Greene K. L., ..., Zietman A. L (2013) **Early Detection of Prostate Cancer: AUA Guideline** *Journal of Urology* **190**:419–426 <https://doi.org/10.1016/j.juro.2013.04.119>
- Chen H. Y., Lin L. T., Wang M. L., Laurent B., Hsu C. H., Pan C. M., Chiou S. H (2017) **Musashi-1 Enhances Glioblastoma Cell Migration and Cytoskeletal Dynamics through Translational Inhibition of Tensin3** *Scientific Reports* **7**:14 <https://doi.org/10.1038/s41598-017-09504-7>
- Christakoudi S., Kakourou A., Markozannes G., Tzoulaki I., Weiderpass E., Brennan P., ..., Tsilidis K. K (2020) **Blood pressure and risk of cancer in the European Prospective Investigation into Cancer and Nutrition** *Int J Cancer* **146**:2680–2693 <https://doi.org/10.1002/ijc.32576>
- Cuchel M., Rader D. J (2006) **Macrophage reverse cholesterol transport: key to the regression of atherosclerosis?** *Circulation* **113**:2548–2555 <https://doi.org/10.1161/circulationaha.104.475715>
- Cui C., Chatterjee B., Lozito T. P., Zhang Z., Francis R. J., Yagi H., ..., Lo C. W (2013) **Wdpcp, a PCP Protein Required for Ciliogenesis, Regulates Directional Cell Migration and Cell Polarity by Direct Modulation of the Actin Cytoskeleton** *Plos Biology* **11**:17 <https://doi.org/10.1371/journal.pbio.1001720>

- Davies N. M., Holmes M. V., Smith G. D (2018) **Reading Mendelian randomisation studies: a guide, glossary, and checklist for clinicians** *Bmj-British Medical Journal* **362**:11 <https://doi.org/10.1136/bmj.k601>
- De Mezer M., Rogalinski J., Przewozny S., Chojnicki M., Niepolski L., Sobieska M., Przystanska A. (2023) **SERPINA3: Stimulator or Inhibitor of Pathological Changes** *Biomedicines* **11**:15 <https://doi.org/10.3390/biomedicines11010156>
- Deichaite I., Sears T. J., Sutton L., Rebibo D., Morgan K., Nelson T., Carter H (2022) **Differential regulation of TNF α and IL-6 expression contributes to immune evasion in prostate cancer** *Journal of Translational Medicine* **20**:16 <https://doi.org/10.1186/s12967-022-03731-x>
- Doyle S. L., Donohoe C. L., Lysaght J., Reynolds J. V (2012) **Visceral obesity, metabolic syndrome, insulin resistance and cancer** *Proc Nutr Soc* **71**:181–189 <https://doi.org/10.1017/s002966511100320x>
- Draisma G., Etzioni R., Tsodikov A., Mariotto A., Wever E., Gulati R., de Koning H. (2009) **Lead Time and Overdiagnosis in Prostate-Specific Antigen Screening: Importance of Methods and Context** *Jnci-Journal of the National Cancer Institute* **101**:374–383 <https://doi.org/10.1093/jnci/djp001>
- Ferkingstad E., Sulem P., Atlason B. A., Sveinbjornsson G., Magnusson M. I., Styrismisdottir E. L., ..., Stefansson K. (2021) **Large-scale integration of the plasma proteome with genetics and disease** *Nature Genetics* **53**:1712 <https://doi.org/10.1038/s41588-021-00978-w>
- Ferlay J., E. M., Lam F., Laversanne M., Colombet M., Mery L, Piñeros M, Znaor A, Soerjomataram I, Bray F. (2024) **Global Cancer Observatory: Cancer Today** <https://gco.iarc.who.int/today>
- Foley C. N., Staley J. R., Breen P. G., Sun B. B., Kirk P. D. W., Burgess S., Howson J. M. M (2021) **A fast and efficient colocalization algorithm for identifying shared genetic risk factors across multiple traits** *Nature Communications* **12**:18 <https://doi.org/10.1038/s41467-020-20885-8>
- Foskolou I. P., Jorgensen C., Leszczynska K. B., Olcina M. M., Tarhonskaya H., Haisma B., ..., Hammond E. M (2017) **Ribonucleotide Reductase Requires Subunit Switching in Hypoxia to Maintain DNA Replication** *Molecular Cell* **66**:206 <https://doi.org/10.1016/j.molcel.2017.03.005>
- Frayling T. M., Colhoun H., Florez J. C (2008) **A genetic link between type 2 diabetes and prostate cancer - Commentary** *Diabetologia* **51**:1757–1760 <https://doi.org/10.1007/s00125-008-1114-9>
- Garde S. V., Basrur V. S., Li L., Finkelman M. A., Krishan A., Wellham L., ..., Tang D. G (1999) **Prostate secretory protein (PSP94) suppresses the growth of androgen-independent prostate cancer cell line (PC3) and xenografts by inducing apoptosis** *Prostate* **38**:118–125 [https://doi.org/10.1002/\(sici\)1097-0045\(19990201\)38:2<118::Aid-pros5>3.0.Co;2-g](https://doi.org/10.1002/(sici)1097-0045(19990201)38:2<118::Aid-pros5>3.0.Co;2-g)
- Giambartolomei C., Vukcevic D., Schadt E. E., Franke L., Hingorani A. D., Wallace C., Plagnol V (2014) **Bayesian Test for Colocalisation between Pairs of Genetic Association Studies Using Summary Statistics** *Plos Genetics* **10**:15 <https://doi.org/10.1371/journal.pgen.1004383>
- Gu Z., Thomas G., Yamashiro J., Shintaku I. P., Dorey F., Raitano A., ..., Reiter R. E (2000) **Prostate stem cell antigen (PSCA) expression increases with high gleason score, advanced stage and bone metastasis in prostate cancer** *Oncogene* **19**:1288–1296 <https://doi.org/10.1038/sj.onc.1203426>

- Hadaschik B. A., Jackson J., Fazli L., Zoubeidi A., Burt H. M., Gleave M. E., So A. I (2008) **Intravesically administered antisense oligonucleotides targeting heat-shock protein-27 inhibit the growth of non-muscle-invasive bladder cancer** *Bju International* **102**:610–616 <https://doi.org/10.1111/j.1464-410X.2008.07669.x>
- Haiman C. A., Stram D. O., Vickers A. J., Wilkens L. R., Braun K., Valtonen-André C., ..., Lilja H. (2013) **Levels of Beta-Microseminoprotein in Blood and Risk of Prostate Cancer in Multiple Populations** *Jnci-Journal of the National Cancer Institute* **105**:237–243 <https://doi.org/10.1093/jnci/djs486>
- Hemani G., Bowden J., Smith G. D (2018) **Evaluating the potential role of pleiotropy in Mendelian randomization studies** *Human Molecular Genetics* **27**:R195–R208 <https://doi.org/10.1093/hmg/ddy163>
- Hobbs B. D., de Jong K., Lamontagne M., Bossé Y., Shrine N., Artigas M. S., ..., Int C. G. C (2017) **Genetic loci associated with chronic obstructive pulmonary disease overlap with loci for lung function and pulmonary fibrosis** *Nature Genetics* **49**:426–432 <https://doi.org/10.1038/ng.3752>
- Hockla A., Miller E., Salameh M. A., Copland J. A., Radisky D. C., Radisky E. S (2012) **PRSS3/Mesotrypsin Is a Therapeutic Target for Metastatic Prostate Cancer** *Molecular Cancer Research* **10**:1555–1566 <https://doi.org/10.1158/1541-7786.Mcr-12-0314>
- Humphries A. C., Mlodzik M (2018) **From instruction to output: Wnt/PCP signaling in development and cancer** *Current Opinion in Cell Biology* **51**:110–116 <https://doi.org/10.1016/j.ceb.2017.12.005>
- Iqbal W., Demidova E. V., Serrao S., ValizadehAslani T., Rosen G., Arora S (2021) **RRM2B Is Frequently Amplified Across Multiple Tumor Types: Implications for DNA Repair, Cellular Survival, and Cancer Therapy** *Frontiers in Genetics* **12**:13 <https://doi.org/10.3389/fgene.2021.628758>
- Ito K., Yamamoto T., Hayashi Y., Sato S., Nakayama J., Urabe F., ..., Yamamoto Y. (2023) **Osteoblast-derived extracellular vesicles exert osteoblastic and tumor-suppressive functions via SERPINA3 and LCN2 in prostate cancer** *Molecular Oncology* **17**:2147–2167 <https://doi.org/10.1002/1878-0261.13484>
- James N. D., Tannock I., N'Dow J., Feng F., Gillessen S., Ali S. A., ..., Xie L.-P. (2024) **The Lancet Commission on prostate cancer: planning for the surge in cases** *Lancet (London, England)* **403**:1683–1722 [https://doi.org/10.1016/s0140-6736\(24\)00651-2](https://doi.org/10.1016/s0140-6736(24)00651-2)
- Jansen B., Zangemeister-Wittke U (2002) **Antisense therapy for cancer - the time of truth** *Lancet Oncology* **3**:672–683 [https://doi.org/10.1016/s1470-2045\(02\)00903-8](https://doi.org/10.1016/s1470-2045(02)00903-8)
- Jiang G. Z., Cao F. Y., Ren G. P., Gao D. L., Bhakta V., Zhang Y. H., ..., Wang Y. H (2010) **PRSS3 promotes tumour growth and metastasis of human pancreatic cancer** *Gut* **59**:1535–1544 <https://doi.org/10.1136/gut.2009.200105>
- Kamada M., So A., Muramaki M., Rocchi P., Beraldi E., Gleave M (2007) **Hsp27 knockdown using nucleotide-based therapies inhibit tumor growth and enhance chemotherapy in human bladder cancer cells** *Molecular Cancer Therapeutics* **6**:299–308 <https://doi.org/10.1158/1535-7163.Mct-06-0417>

- Kaufman K. D (1996) **Androgen metabolism as it affects hair growth in androgenetic alopecia** *Dermatol Clin* **14**:697–711 [https://doi.org/10.1016/s0733-8635\(05\)70396-x](https://doi.org/10.1016/s0733-8635(05)70396-x)
- Lawrence M. G., Lai J., Clements J. A (2010) **Kallikreins on steroids: structure, function, and hormonal regulation of prostate-specific antigen and the extended kallikrein locus** *Endocr Rev* **31**:407–446 <https://doi.org/10.1210/er.2009-0034>
- Li E. M., Liu L. H., Li F. T., Luo L. M., Zhao S. K., Wang J. M., ..., Zhao Z. G (2017) **PSCA promotes prostate cancer proliferation and cell-cycle progression by up-regulating c-Myc** *Prostate* **77**:1563–1572 <https://doi.org/10.1002/pros.23432>
- Liang Z., Xie B., Li J., Wang X., Wang S., Meng S., ..., Xie L. (2016) **Hypertension and risk of prostate cancer: a systematic review and meta-analysis** *Scientific Reports* **6**:31358 <https://doi.org/10.1038/srep31358>
- Lilja H., Ulmert D., Vickers A. J (2008) **Prostate-specific antigen and prostate cancer: prediction, detection and monitoring** *Nat Rev Cancer* **8**:268–278 <https://doi.org/10.1038/nrc2351>
- Machiela M. J., Lindström S., Allen N. E., Haiman C. A., Albanes D., Barricarte A., ..., Kraft P. (2012) **Association of Type 2 Diabetes Susceptibility Variants With Advanced Prostate Cancer Risk in the Breast and Prostate Cancer Cohort Consortium** *American Journal of Epidemiology* **176**:1121–1129 <https://doi.org/10.1093/aje/kws191>
- Mantovani A., Allavena P., Sica A., Balkwill F (2008) **Cancer-related inflammation** *Nature* **454**:436–444 <https://doi.org/10.1038/nature07205>
- Martuszevska D., Ljungberg B., Johansson M., Landberg G., Oslakovic C., Dahlbäck B., Hafizi S (2009) **Tensin3 Is a Negative Regulator of Cell Migration and All Four Tensin Family Members Are Downregulated in Human Kidney Cancer** *Plos One* **4**:13 <https://doi.org/10.1371/journal.pone.0004350>
- Mavridis K., Avgeris M., Scorilas A (2014) **Targeting kallikrein-related peptidases in prostate cancer** *Expert Opin Ther Targets* **18**:365–383 <https://doi.org/10.1517/14728222.2014.880693>
- Moyad M. A (2015) **Preventing aggressive prostate cancer with proven cardiovascular disease preventive methods** *Asian J Androl* **17**:874–877 <https://doi.org/10.4103/1008-682x.156854>
- Nakajima T., Fujino S., Nakanishi G., Kim Y. S., Jetten A. M (2004) **TIP27: a novel repressor of the nuclear orphan receptor TAK1/TR4** *Nucleic Acids Research* **32**:4194–4204 <https://doi.org/10.1093/nar/gkh741>
- Nam R. K., Reeves J. R., Toi A., Dulude H., Trachtenberg J., Emami M., ..., Narod S. A (2006) **A novel serum marker, total prostate secretory protein of 94 amino acids, improves prostate cancer detection and helps identify high grade cancers at diagnosis** *Journal of Urology* **175**:1291–1297 [https://doi.org/10.1016/s0022-5347\(05\)00695-6](https://doi.org/10.1016/s0022-5347(05)00695-6)
- Ochoa D., Hercules A., Carmona M., Suveges D., Baker J., Malangone C., ..., McDonagh E. (2023) **The next-generation Open Targets Platform: reimaged, redesigned, rebuilt** *Nucleic Acids Research* **51**:D1353–D1359 <https://doi.org/10.1093/nar/gkac1046>
- Oesterling J. E (1991) **Prostate Specific Antigen: A Critical Assessment of the Most Useful Tumor Marker for Adenocarcinoma of the Prostate** *Journal of Urology* **145**:907–923 <https://doi.org/10.1093/ajur/145.5.907>

[doi.org/10.1016/s0022-5347\(17\)38491-4](https://doi.org/10.1016/s0022-5347(17)38491-4)

Okumura H., Natsugoe S., Matsumoto M., Mataka Y., Takatori H., Ishigami S., ..., Aikou T. (2005) **The predictive value of p53, p53R2, and p21 for the effect of chemoradiation therapy on oesophageal squamous cell carcinoma** *British Journal of Cancer* **92**:284–289 <https://doi.org/10.1038/sj.bjc.6602322>

Okuno M., Adachi S., Kozawa O., Shimizu M., Yasuda I (2016) **The Clinical Significance of Phosphorylated Heat Shock Protein 27 (HSPB1) in Pancreatic Cancer** *International Journal of Molecular Sciences* **17**:7 <https://doi.org/10.3390/ijms17010137>

Ossoli A., Wolska A., Remaley A. T., Gomaschi M (2022) **High-density lipoproteins: A promising tool against cancer** *Biochim Biophys Acta Mol Cell Biol Lipids* **1867**:159068 <https://doi.org/10.1016/j.bbalip.2021.159068>

Perez-Cornago A., Dunneram Y., Watts E. L., Key T. J., Travis R. C (2022) **Adiposity and risk of prostate cancer death: a prospective analysis in UK Biobank and meta-analysis of published studies** *Bmc Medicine* **20**:13 <https://doi.org/10.1186/s12916-022-02336-x>

Petrelli F., Cortellini A., Indini A., Tomasello G., Ghidini M., Nigro O., ..., Zaniboni A. (2021) **Association of Obesity With Survival Outcomes in Patients With Cancer A Systematic Review and Meta-analysis** *Jama Network Open* **4**:30 <https://doi.org/10.1001/jamanetworkopen.2021.3520>

Pietzner M., Wheeler E., Carrasco-Zanini J., Cortes A., Koprulu M., Wörheide M. A., ..., Langenberg C. (2021) **Mapping the proteo-genomic convergence of human diseases** *Science* **374**:839 <https://doi.org/10.1126/science.abj1541>

Reeves J. R., Dulude H., Panchal C., Daigneault L., Ramnani D. M (2006) **Prognostic value of prostate secretory protein of 94 amino acids and its binding protein after radical prostatectomy** *Clinical Cancer Research* **12**:6018–6022 <https://doi.org/10.1158/1078-0432.Ccr-06-0625>

Rohatgi A., Westterterp M., von Eckardstein A., Remaley A., Rye K. A. (2021) **HDL in the 21st Century: A Multifunctional Roadmap for Future HDL Research** *Circulation* **143**:2293–2309 <https://doi.org/10.1161/circulationaha.120.044221>

Rosario S. R., Jacobi J. J., Long M. D., Affronti H. C., Rowsam A. M., Smiraglia D. J (2023) **JAZF1: A Metabolic Regulator of Sensitivity to a Polyamine-Targeted Therapy** *Molecular Cancer Research* **21**:24–35 <https://doi.org/10.1158/1541-7786.Mcr-22-0316>

Ruscica M., Botta M., Ferri N., Giorgio E., Macchi C., Franceschini G., ..., Gomaschi M. (2018) **High Density Lipoproteins Inhibit Oxidative Stress-Induced Prostate Cancer Cell Proliferation** *Scientific Reports* **8**:2236 <https://doi.org/10.1038/s41598-018-19568-8>

Saxena R., Voight B. F., Zeggini E., Scott L. J., Genetics I. C. D (2008) **Meta-analysis of genome-wide association data and large-scale replication identifies several additional susceptibility loci for type 2 diabetes** *Diabetes* **57**:A41–A41

Schumacher F. R., Al Olama A. A., Berndt S. I., Benlloch S., Ahmed M., Saunders E. J., ..., Elucidating Loci Involved P. (2018) **Association analyses of more than 140,000 men identify**

63 new prostate cancer susceptibility loci *Nature Genetics* **51**:363–363 <https://doi.org/10.1038/s41588-018-0330-6>

Seretis A., Cividini S., Markozannes G., Tseretopoulou X., Lopez D. S., Ntzani E. E., Tsilidis K. K. (2019) **Association between blood pressure and risk of cancer development: a systematic review and meta-analysis of observational studies** *Scientific Reports* **9**:8565 <https://doi.org/10.1038/s41598-019-45014-4>

Sheng B. W., Qi C. C., Liu B., Lin Y., Fu T., Zeng Q. D. (2017) **Increased HSP27 correlates with malignant biological behavior of non-small cell lung cancer and predicts patient's survival** *Scientific Reports* **7**:12 <https://doi.org/10.1038/s41598-017-13956-2>

Shi D. B., Ma R. R., Zhang H., Hou F., Guo X. Y., Gao P. (2019) **GAGE7B promotes tumor metastasis and growth via activating the p38/pMAPKAPK2/pHSP27 pathway in gastric cancer** *Journal of Experimental & Clinical Cancer Research* **38**:12 <https://doi.org/10.1186/s13046-019-1125-z>

Shiota M., Bishop J. L., Nip K. M., Zardan A., Takeuchi A., Cordonnier T., ..., Zoubeidi A. (2013) **Hsp27 Regulates Epithelial Mesenchymal Transition, Metastasis, and Circulating Tumor Cells in Prostate Cancer** *Cancer Research* **73**:3109–3119 <https://doi.org/10.1158/0008-5472.Can-12-3979>

Stikbakke E., Schirmer H., Knutsen T., Støyten M., Wilsgaard T., Giovannucci E. L., ..., Thune I. (2022) **Systolic and diastolic blood pressure, prostate cancer risk, treatment, and survival. The PROCA-life study** *Cancer Med* **11**:1005–1015 <https://doi.org/10.1002/cam4.4523>

Sun B. B., Chiou J., Traylor M., Benner C., Hsu Y. H., Richardson T. G., ..., Whelan C. D. (2023) **Plasma proteomic associations with genetics and health in the UK Biobank** *Nature* **622**:329–338 <https://doi.org/10.1038/s41586-023-06592-6>

Sung Y., Park S., Park S. J., Jeong J., Choi M., Lee J., ..., Kim M. O. (2018) **Jazf1 promotes prostate cancer progression by activating JNK/Slug** *Oncotarget* **9**:755–765 <https://doi.org/10.18632/oncotarget.23146>

Teng K. Y., Mansour A. G., Zhu Z., Li Z. Y., Tian L., Ma S. B., ..., Yu J. H. (2022) **Off-the-Shelf Prostate Stem Cell Antigen-Directed Chimeric Antigen Receptor Natural Killer Cell Therapy to Treat Pancreatic Cancer** *Gastroenterology* **162**:1319–1333 <https://doi.org/10.1053/j.gastro.2021.12.281>

Thomas G., Jacobs K. B., Yeager M., Kraft P., Wacholder S., Orr N., ..., Chanock S. J. (2008) **Multiple loci identified in a genome-wide association study of prostate cancer** *Nature Genetics* **40**:310–315 <https://doi.org/10.1038/ng.91>

Tong Y., Guo Y. J., Zhang Q., Bi H. X., Kai K., Zhou R. Y. (2022) **Combined treatment with dihydrotestosterone and lipopolysaccharide modulates prostate homeostasis by upregulating TNF- α from M1 macrophages and promotes proliferation of prostate stromal cells** *Asian J Androl* **24**:513–520 <https://doi.org/10.4103/aja2021114>

Vasiljevic N., Ahmad A. S., Beesley C., Thorat M. A., Fisher G., Berney D. M., ..., Lorincz A. T. (2013) **Association between DNA methylation of HSPB1 and death in low Gleason score prostate cancer** *Prostate Cancer and Prostatic Diseases* **16**:35–40 <https://doi.org/10.1038/pcan.2012.47>

- Wain L. V., Shrine N., Artigas M. S., Erzurumluoglu A. M., Noyvert B., Bossini-Castillo L., ..., Geisinger-Regeneron Discov E. H. R (2017) **Genome-wide association analyses for lung function and chronic obstructive pulmonary disease identify new loci and potential druggable targets** *Nature Genetics* **49**:416–425 <https://doi.org/10.1038/ng.3787>
- Wang L., Balmat T. J., Antonia A. L., Constantine F. J., Henao R., Burke T. W., ..., Ko D. C. (2020) **An atlas connecting shared genetic architecture of human diseases and molecular phenotypes provides insight into COVID-19 susceptibility** *medRxiv* <https://doi.org/10.1101/2020.12.20.20248572>
- Wang X. J., Su H. B (2010) **Unraveling Enigma in the Z-Disk** *Circulation Research* **107**:321–323 <https://doi.org/10.1161/circresaha.110.225615>
- Watanabe K., Taskesen E., van Bochoven A., Posthuma D. (2017) **Functional mapping and annotation of genetic associations with FUMA** *Nature Communications* **8**:11 <https://doi.org/10.1038/s41467-017-01261-5>
- Xie P. B., Liu M. S., Chen F., Wu S. M., Shao T., Wang W., ..., Zhou H. Q. (2020) **Long Non-coding RNA AGAP2-AS1 Silencing Inhibits PDLIM5 Expression Impeding Prostate Cancer Progression via Up-Regulation of MicroRNA-195-5p** *Frontiers in Genetics* **11**:13 <https://doi.org/10.3389/fgene.2020.01030>
- Xing C. J., Ye H., Wang W. W., Sun M. M., Zhang J. Y., Zhao Z. H., Jiang G. Z (2019) **Circular RNA ADAM9 facilitates the malignant behaviours of pancreatic cancer by sponging miR-217 and upregulating PRSS3 expression** *Artificial Cells Nanomedicine and Biotechnology* **47**:3920–3928 <https://doi.org/10.1080/21691401.2019.1671856>
- Xing Z. S., Li S. L., Liu Z. X., Zhang C., Bai Z. M (2021) **CircSERPINA3 regulates SERPINA3-mediated apoptosis, autophagy and aerobic glycolysis of prostate cancer cells by competitively binding to MiR-653-5p and recruiting BUD13** *Journal of Translational Medicine* **19**:13 <https://doi.org/10.1186/s12967-021-03063-2>
- Zheng J. Y., Wang C. Y., Gao C., Xiao Q., Huang C. W., Wu M., Li L. Y (2021) **MLL3 suppresses tumorigenesis through regulating TNS3 enhancer activity** *Cell Death & Disease* **12**:13 <https://doi.org/10.1038/s41419-021-03647-2>
- Zhigang Z., Wenlv S (2004) **Prostate stem cell antigen (PSCA) expression in human prostate cancer tissues and its potential role in prostate carcinogenesis and progression of prostate cancer** *World journal of surgical oncology* **2**:13 <https://doi.org/10.1186/1477-7819-2-13>
- Zhou M. Y., Li K. K., Luo K. Q (2023) **Shear Stress Drives the Cleavage Activation of Protease-Activated Receptor 2 by PRSS3/Mesotrypsin to Promote Invasion and Metastasis of Circulating Lung Cancer Cells** *Advanced Science* **10**:21 <https://doi.org/10.1002/advs.202301059>
- Zuidema A., Atherton P., Kreft M., Hoekman L., Bleijerveld O. B., Nagaraj N., ..., Sonnenberg A. (2022) **PEAK1 Y635 phosphorylation regulates cell migration through association with Tensin3 and integrins** *Journal of Cell Biology* **221**:18 <https://doi.org/10.1083/jcb.202108027>

Author information

Chen Lin[#]

Department of Pharmacy, Peking University First Hospital, Beijing, China

[#]Contributed equally.

Rong Zhuona[#]

Department of Pharmacy, Peking University First Hospital, Beijing, China

[#]Contributed equally.

Gu Yanlun

School of Pharmaceutical Sciences, Peking University, Beijing, China

Chen Yuke

Department of Urology Surgery, Peking University Third Hospital, Beijing, China

Yu Wei

Department of Urology Surgery, Peking University Third Hospital, Beijing, China

Zhou Ying

Department of Pharmacy, Peking University First Hospital, Beijing, China

For correspondence: zhouying0321@126.com

Pang Xiaocong

Department of Pharmacy, Peking University First Hospital, Beijing, China

ORCID iD: [0000-0002-5951-5944](https://orcid.org/0000-0002-5951-5944)

For correspondence: pangxiaocong1227@163.com

Editors

Reviewing Editor

Jennifer Cullen

Case Western Reserve University, Cleveland, United States of America

Senior Editor

Eduardo Franco

McGill University, Montreal, Canada

Reviewer #1 (Public review):

Summary:

In Causal associations between plasma proteins and prostate cancer: a Proteome-Wide Mendelian Randomization, the authors present a manuscript which seeks to identify novel markers for prostate cancer through analysis of large biobank-based datasets and to extend

this analysis to potential therapeutic targets for drugs. This is an area that is already extensively researched, but remains important, due to the high burden and mortality of prostate cancer globally.

Strengths:

The main strengths of the manuscript are the identification and use of large biobank data assets, which provide large numbers of cases and controls, essential for achieving statistical power. The databases used (deCODE, FinnGen, and the UK Biobank) allow for robust numbers of cases and controls. The analytical method chosen, Mendelian Randomization, is appropriate to the problem. Another strength is the integration of multi-omic datasets, here using protein data as well as GWAS sources to integrate genomic and proteomic data.

Weaknesses:

The main weaknesses of the manuscript relate to the following areas:

(1) The failure of the study to analyse the data in the context of other closely related conditions such as benign prostatic hyperplasia (BPH) or lower urinary tract symptoms (LUTS), which have some pathways and biomarkers in common, such as inflammatory pathways (including complement) and specific markers such as KLK3. As a consequence, it is not possible for readers to know whether the findings are specific to prostate cancer or whether they are generic to prostate dysfunction. Given the prevalence of prostate dysfunction (half of men reaching their sixth decade), the potential for false positives and overtreatment from non-specific biomarkers is a major problem, resulting in the evidence presented in this manuscript being weak. Other researchers have addressed this issue using the same data sources as presented here, for example, in this paper, looking at BPH in the UK Biobank population.

<https://www.nature.com/articles/s41467-018-06920-9>

(2) There is no discussion of Gleason scores with regard to either biomarkers or therapies, and a general lack of discussion around indolent disease as compared with more aggressive variants. These are crucial issues with regard to the triage and identification of genomically aggressive localized prostate cancers. See, for example, the work set out in: <https://doi.org/10.1038/nature20788>.

(3) An additional issue is that the field of PCa research is fast-moving. The manuscript cites ~80 references, but too few of these are from recent studies, and many important and relevant papers are not included. The manuscript would be much stronger if it compared and contrasted its findings with more recent studies of PCa biomarkers and targets, especially those concerned with multi-omics and those including BPH.

(4) The Methods section provides no information on how the Controls were selected. There is no Table providing cohort data to allow the reader to know whether there were differences in age, BMI, ethnic grouping, social status or deprivation, or smoking status, between the Cases and Controls. These types of data are generally recorded in Biobank data, so this sort of analysis should be possible, or if not, the authors' inability to construct an appropriately matched set of Controls should be discussed as a Limitation.

Assessing impact:

Because of the weaknesses of the approach identified above, without further additions to the manuscript, the likely impact of the work on the field is minimal. There is no significant utility of the methods and data to the community, because the data are pre-existing and are not newly introduced to the community in this work, and Mendelian randomization is a well-described approach in common use, and therefore, the assets and methods described in the manuscript are not novel. With regard to the authors achieving their aims, without assessing

specificity and without setting their findings in the context of the latest literature, the authors (and readers) cannot know or assess whether the biomarkers identified or the druggable targets will be useful in the clinic.

In conclusion, adding additional context and analysis to the manuscript would both help readers interpret and understand the work and would also greatly enhance its significance. For example, the UK Biobank includes data on men with BPH / LUTS, as analysed in this paper, for example, <https://doi.org/10.1038/s41467-018-06920-9>. By extending this analysis to identify which biomarkers and druggable targets are specific to PCa, and which are generic to prostate dysfunction, the authors would substantially reduce the risks of diagnostic false positives. This would help to manage the risks of inappropriate treatment or overtreatment.

<https://doi.org/10.7554/eLife.101584.1.sa2>

Reviewer #2 (Public review):

This is potentially interesting work, but the analyses are attempted in a rather scattergun way, with little evident critical thought. The structure of the work (Results before Methods) can work in some manuscripts, but it is not ideal here. The authors discuss results before we know anything about the underlying data that the results come from. It gives the impression that the authors regard data as a resource to be exploited, without really caring where the data comes from. The methods can provide meaningful insights if correctly used, but while I don't have reasons to doubt that the analyses were conducted correctly, findings are presented with little discussion or interpretation. No follow-up analyses are performed.

In summary, there are likely some gems here, but the whole manuscript is essentially the output from an analytic pipeline.

Taking the researchers aims in turn:

(1) Meta-GWAS - while combining two datasets together can provide additional insights, the contribution of this analysis above existing GWAS is not clear. The PRACTICAL consortium has already reported the GWAS of 70% of these data. What additional value does this analysis provide? (Likely some, but it's not clear from the text.) Also, the presentation of results is unclear - authors state that only 5 gene regions contained variants at $p < 5 \times 10^{-8}$, but Figure 1 shows dozens of hits above 5×10^{-8} . Also, the red line in Figure 1 (supposedly at 5×10^{-8}) is misplaced.

(2) Cross-phenotype analysis. It is not really clear what this analysis is, or why it is done. What is the iCPAGdb? A database? A statistical method? Why would we want to know cross-phenotype associations? What even are these? It seems that the authors have taken data from an online resource and have written a paragraph based on this existing data with little added value.

(3) PW-MR. I can see the value of this work, but many details are unclear. Was this a two-sample MR using PRACTICAL + FinnGen data for the outcome? How many variants were used in key analyses? Again, the description of results is sparse and gives little added value.

(4) Colocalization - seems clear to me.

(5) Additional post-GWAS analyses (pathway + druggability) - again, the analyses seem to be performed appropriately, although little additional insight other than the reporting of output from the methods.

Minor points:

(6) The stated motivation for this work is "early detection". But causality isn't necessary for early detection. If the authors are interested in early detection, other analysis approaches are more appropriate.

(7) The authors state "193 proteins were associated with PCa risk", but they are looking at MR results - these analyses test for disease associations of genetically-predicted levels of proteins, not proteins themselves.

Strengths:

The data and methods used are state-of-the-art.

Weaknesses:

The reader will have to provide their own translational insight.

<https://doi.org/10.7554/eLife.101584.1.sa1>

Author response:

Public Reviews:

Reviewer #1 (Public review):

Summary:

In Causal associations between plasma proteins and prostate cancer: a Proteome-Wide Mendelian Randomization, the authors present a manuscript which seeks to identify novel markers for prostate cancer through analysis of large biobank-based datasets and to extend this analysis to potential therapeutic targets for drugs. This is an area that is already extensively researched, but remains important, due to the high burden and mortality of prostate cancer globally.

Strengths:

The main strengths of the manuscript are the identification and use of large biobank data assets, which provide large numbers of cases and controls, essential for achieving statistical power. The databases used (deCODE, FinnGen, and the UK Biobank) allow for robust numbers of cases and controls. The analytical method chosen, Mendelian Randomization, is appropriate to the problem. Another strength is the integration of multi-omic datasets, here using protein data as well as GWAS sources to integrate genomic and proteomic data.

Thank you for your positive feedback regarding the overall quality of our work and we greatly appreciate you taking time and making effort in reviewing our manuscript.

Weaknesses:

The main weaknesses of the manuscript relate to the following areas:

(1) The failure of the study to analyse the data in the context of other closely related conditions such as benign prostatic hyperplasia (BPH) or lower urinary tract symptoms (LUTS), which have some pathways and biomarkers in common, such as inflammatory pathways (including complement) and specific markers such as KLK3. As a consequence, it is not possible for readers to know whether the findings are specific to prostate cancer or whether they are generic to prostate dysfunction. Given the prevalence of prostate

dysfunction (half of men reaching their sixth decade), the potential for false positives and overtreatment from non-specific biomarkers is a major problem, resulting in the evidence presented in this manuscript being weak. Other researchers have addressed this issue using the same data sources as presented here, for example, in this paper, looking at BPH in the UK Biobank population. <https://www.nature.com/articles/s41467-018-06920-9>

Thank you for your valuable comment. We fully agree that biomarker development must prioritize specificity to avoid overtreatment. While our study is a foundational step toward identifying potential therapeutic targets or complementary biomarkers for prostate cancer (PCa)—not as a direct endorsement of these proteins for standalone clinical diagnosis. Mendelian randomization (MR) analysis strengthens causal inference by design, and we further ensured robustness through sensitivity analyses (e.g. MR-Egger regression for pleiotropy, Bonferroni correction for multiple testing). These methods distinguish true causal effects from nonspecific associations. Importantly, while PSA's lack of specificity is widely recognized, its role in reducing PCa mortality underscores the value of biomarker-driven screening. Our findings align with the need to integrate multiple markers (e.g. combining a novel protein with PSA) to improve diagnostic precision. Translating these causal insights into clinical tools remains challenging but represents a necessary next step, and we emphasize that this work provides a rigorous starting point for future validation studies.

(2) There is no discussion of Gleason scores with regard to either biomarkers or therapies, and a general lack of discussion around indolent disease as compared with more aggressive variants. These are crucial issues with regard to the triage and identification of genomically aggressive localized prostate cancers. See, for example, the work set out in: <https://doi.org/10.1038/nature20788>

Thank you for pointing this out. We acknowledge that our original analysis did not directly address this critical issue due to a key data limitation: the publicly available GWAS summary statistics for PCa (from openGWAS and FinnGen) do not provide genetic associations stratified by phenotypic severity or molecular subtypes. This limitation precluded MR analysis of proteins specifically linked to aggressive disease. To partially bridge this gap, we integrate evidence from recent studies in the revised Discussion section to explore the relevance of potential biomarkers to aggressive PCa.

(3) An additional issue is that the field of PCa research is fast-moving. The manuscript cites ~80 references, but too few of these are from recent studies, and many important and relevant papers are not included. The manuscript would be much stronger if it compared and contrasted its findings with more recent studies of PCa biomarkers and targets, especially those concerned with multi-omics and those including BPH.

Thank you for your professional comments. We have rigorously updated the manuscript to include more recent publications and we systematically compare and contrast our findings with these recent studies in the revised Discussion section.

(4) The Methods section provides no information on how the Controls were selected. There is no Table providing cohort data to allow the reader to know whether there were differences in age, BMI, ethnic grouping, social status or deprivation, or smoking status, between the Cases and Controls. These types of data are generally recorded in Biobank data, so this sort of analysis should be possible, or if not, the authors' inability to construct an appropriately matched set of Controls should be discussed as a Limitation.

We thank the reviewer for raising this important methodological concern. We have expanded the Limitations section to state it.

Reviewer #2 (Public review):

This is potentially interesting work, but the analyses are attempted in a rather scattergun way, with little evident critical thought. The structure of the work (Results before Methods) can work in some manuscripts, but it is not ideal here. The authors discuss results before we know anything about the underlying data that the results come from. It gives the impression that the authors regard data as a resource to be exploited, without really caring where the data comes from. The methods can provide meaningful insights if correctly used, but while I don't have reasons to doubt that the analyses were conducted correctly, findings are presented with little discussion or interpretation. No follow-up analyses are performed.

In summary, there are likely some gems here, but the whole manuscript is essentially the output from an analytic pipeline.

We thank the reviewer for the thoughtful evaluation of our work.

Taking the researchers aims in turn:

(1) Meta-GWAS - while combining two datasets together can provide additional insights, the contribution of this analysis above existing GWAS is not clear. The PRACTICAL consortium has already reported the GWAS of 70% of these data. What additional value does this analysis provide? (Likely some, but it's not clear from the text.) Also, the presentation of results is unclear - authors state that only 5 gene regions contained variants at $p < 5 \times 10^{-8}$, but Figure 1 shows dozens of hits above 5×10^{-8} . Also, the red line in Figure 1 (supposedly at 5×10^{-8}) is misplaced.

Thank you very much for your feedback. Although the PRACTICAL consortium constituted the majority of PCa GWAS data, our meta-analysis integrating FinnGen data enhanced statistical power enabling robust detection of low-frequency variants with minor allele frequencies. Moreover, FinnGen's Finnish ancestry (genetic isolate) helps distinguish population-specific effects. The presentation of results showed the top 5 gene regions contained variants at $p < 5 \times 10^{-8}$. We apologize for not noticing that the red line was not displayed correctly in the original figures included in the manuscript. We have updated it in the revised manuscript.

(2) Cross-phenotype analysis. It is not really clear what this analysis is, or why it is done. What is the iCPAGdb? A database? A statistical method? Why would we want to know cross-phenotype associations? What even are these? It seems that the authors have taken data from an online resource and have written a paragraph based on this existing data with little added value.

We thank you for raising this issue. The iCPAGdb (interactive Cross-Phenotype Analysis of GWAS database) is an integrative platform that systematically identifies cross-phenotype associations and evaluates genetic pleiotropy by leveraging LD-proxy associations from the NHGRI-EBI GWAS Catalog. The pathogenesis and progression of prostate cancer constitute a complex pathophysiological continuum characterized by dynamic multisystem interactions, extending beyond singular molecular pathway dysregulation to encompass coordinated disruptions across endocrine regulation, immune microenvironment remodeling, and metabolic reprogramming. Therefore, it is indispensable for discriminating primary pathogenic drivers from secondary compensatory responses, ultimately informing the development of precision therapeutic strategies.

(3) PW-MR. I can see the value of this work, but many details are unclear. Was this a two-sample MR using PRACTICAL + FinnGen data for the outcome? How many variants were used in key analyses? Again, the description of results is sparse and gives little added value.

We thank you for raising this issue. Two-sample MR refers to an analytical design where genetic instruments for the exposure (plasma proteins) and genetic associations with the outcome (PCa) are derived from non-overlapping populations. This ensures complete sample independence between exposure and outcome datasets to avoid confounding biases, regardless of whether the outcome data originate from single or multiple cohorts. The meta-analysis of PRACTICAL and FinnGen GWAS generates 27,210 quality-controlled variants ($p < 5 \times 10^{-8}$, $MAF \geq 1\%$, LD-clumped $r^2 < 0.1$) used in key analyses.

(4) Colocalization - seems clear to me.

(5) Additional post-GWAS analyses (pathway + druggability) - again, the analyses seem to be performed appropriately, although little additional insight other than the reporting of output from the methods.

The post-MR druggability and pathway analyses serve two primary scientific purposes: (1) therapeutic prioritization - systematically evaluating which MR-identified proteins represent tractable drug targets (either through existing FDA-approved agents or compounds in clinical development) with direct relevance to cancer or PCa management, and (2) mechanistic hypothesis generation - mapping these candidate proteins to coherent biological pathways to guide future functional validation studies investigating their causal roles in prostate carcinogenesis.

Minor points:

(6) The stated motivation for this work is "early detection". But causality isn't necessary for early detection. If the authors are interested in early detection, other analysis approaches are more appropriate.

We appreciate your insightful feedback. While early detection is one motivation for this work, our primary goal extends to identifying causally implicated proteins that may serve as intervention targets for PCa prevention or therapy. Establishing causality is critical for distinguishing biomarkers that drive disease pathogenesis from those that are secondary to disease progression, as the former holds greater specificity for early detection and prioritization of therapeutic targets. While we acknowledge that validation for early detection may require additional methodologies, MR analysis provides a foundational step by prioritizing candidate proteins with causal links to disease. This approach ensures that downstream efforts focus on biomarkers and targets with the greatest potential to alter disease trajectories, rather than merely correlative markers.

(7) The authors state "193 proteins were associated with PCa risk", but they are looking at MR results - these analyses test for disease associations of genetically-predicted levels of proteins, not proteins themselves.

In MR, the exposure of interest is the lifelong effect of genetically predicted protein levels. This approach is designed to infer causality while avoiding confounding and reverse causation, as genetic variants are fixed at conception and unaffected by disease processes. When we state "193 proteins were associated with PCa risk," we specifically refer to proteins whose genetically predicted levels (based on instrument SNPs from protein QTLs) show causal links to PCa. Importantly, MR does not measure the direct association between

observed protein concentrations and disease. Instead, it estimates the lifelong causal effect of protein levels predicted by genetics. This distinction is critical for disentangling cause from consequence. For example, a protein elevated due to tumor progression would not be identified as causal in MR if its genetic predictors are unrelated to PCa risk.

We acknowledge that clinical translation requires further validation of these proteins in observational studies measuring actual protein levels. However, MR provides a robust first step by prioritizing candidates with causal roles, thereby reducing the risk of investing in biomarkers confounded by disease processes.

<https://doi.org/10.7554/eLife.101584.1.sa0>